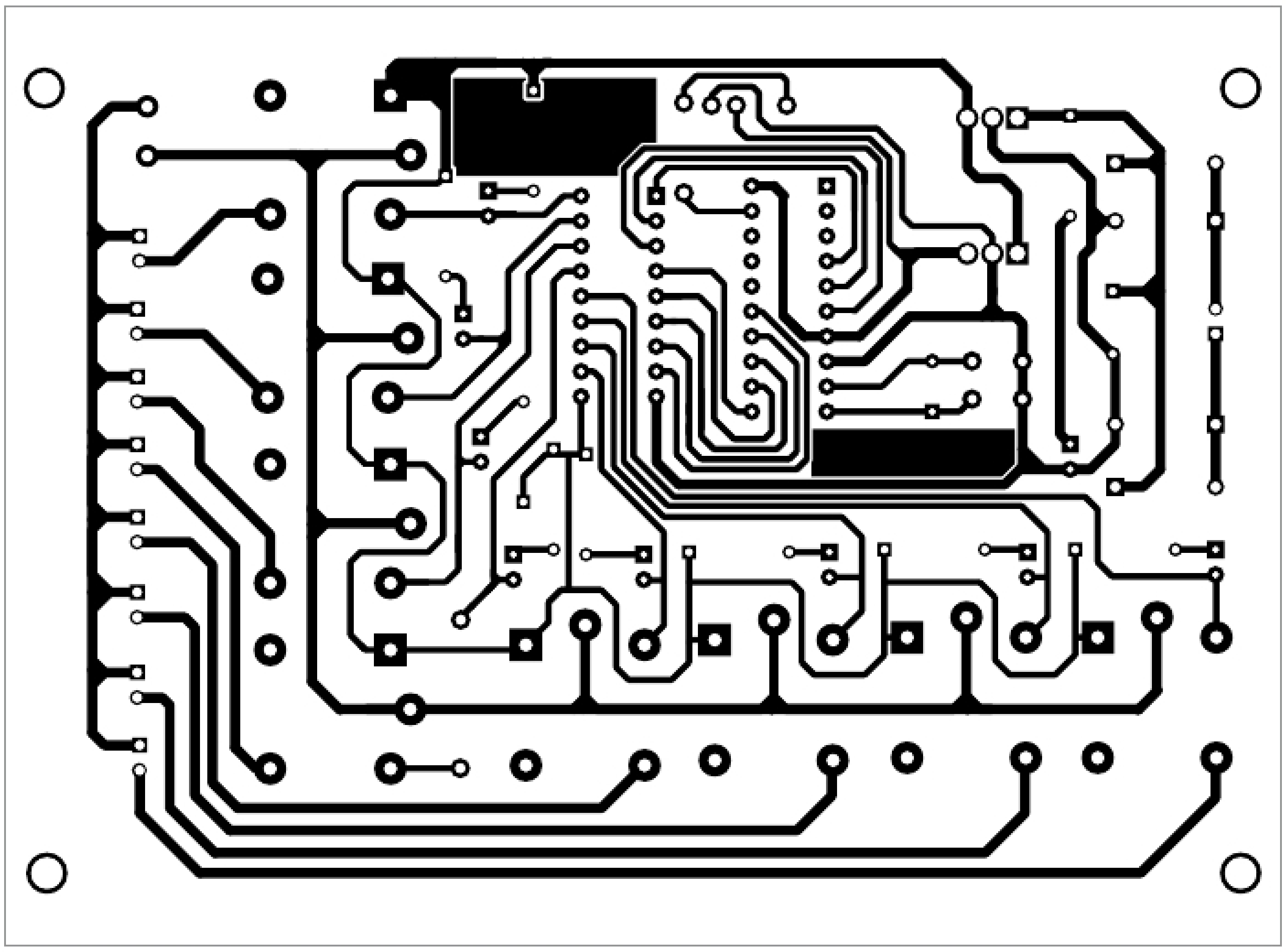

Fig. 6: An actual size PCB layout for the device controller

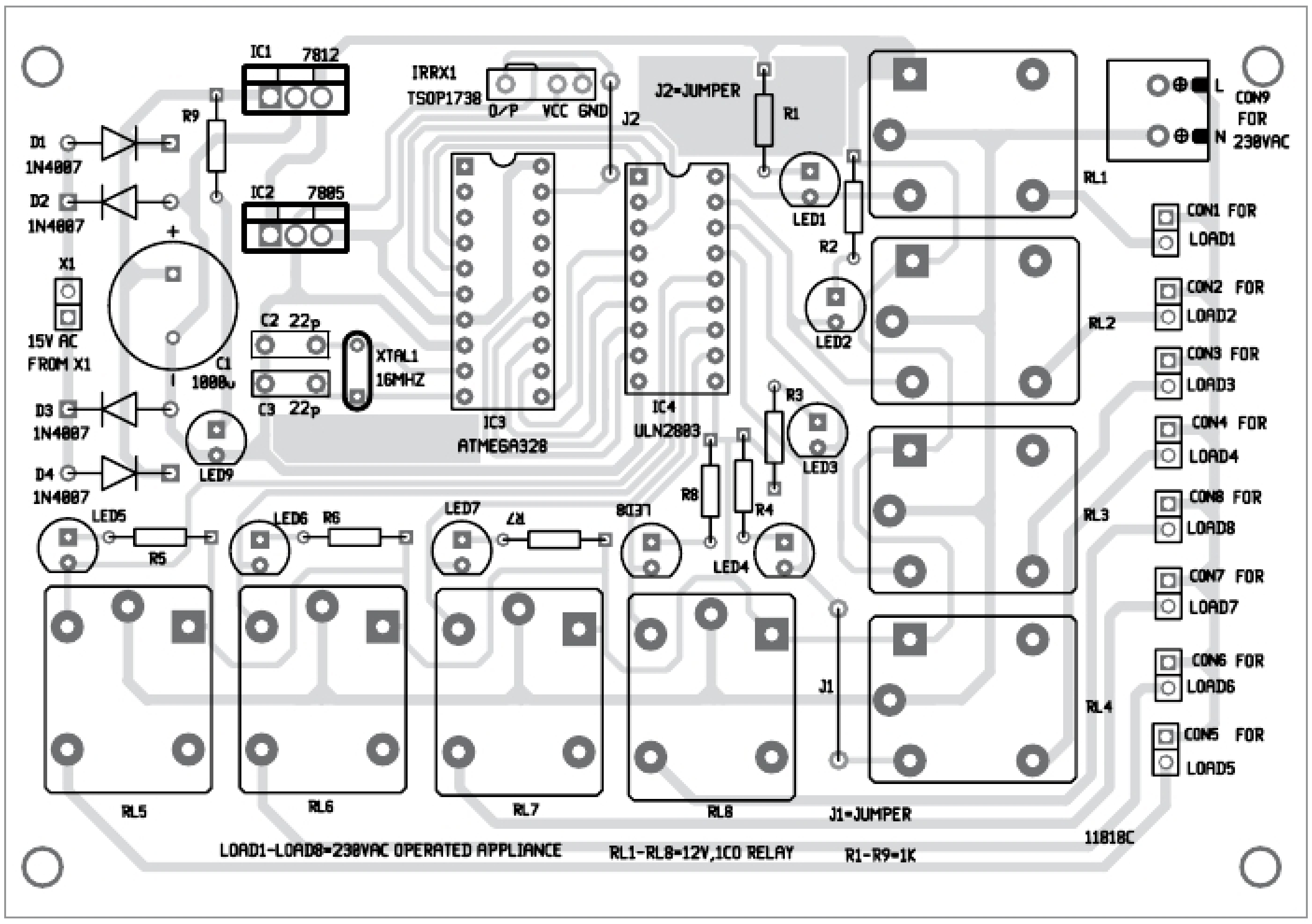

Fig. 7: Components layout of the PCB

during testing are listed in Table 1. You should use these decoded values in your final Arduino code (devicecontefylab.ino). After entering the decoded value of each key in the final code, compile it and upload it to the Arduino Uno board. Then remove the ATemga328P MCU from Arduino Uno board and use it in the main device control circuit as shown in Fig. 3.

Relay driver and power isolator unit. This section includes ULN2803 (IC4) relay driver, which is used to drive the relays as well as to isolate low-voltage section of the circuit from high-voltage section. Basically, it is used for switching on or switching off a high-voltage device connected through a set of relay contacts.

Construction and testing

An actual-size PCB layout for device controller is shown in Fig. 6 and its components layout in Fig. 7.

First, decode each key of the remote control, include decoded values in devicecontefylab.ino code and upload it to the MCU as explained earlier. Remove the MCU from Arduino board and insert it into the PCB. It is recommended to use a 28-pin IC socket for the MCU.

Connect eight loads across Load1 through Load8. Connect secondary of transformer X1 to the PCB for power supply. Point the IR remote towards TSOP1738 module from a distance of about one metre.

When you press a button of the remote, say key 1, relay RL1 energises and Load1 turns on. When you press power on/off key of remote control, relay RL1 de-energises and Load1 turns off. Similarly, you can operate the remaining loads by pressing remote buttons 2 through 8 as listed in the Table. EFY

EFY Note

The source code of this project is available for free download at *source.efymag.com*

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